

Report
Of
SUMMER INTERNSHIP

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of: 2410030568
Name of Student: ANJALI
Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, ANJALI, do hereby declare that the internship report titled "Software Development Intern at Naadyx Solutions Private Limited (Formerly Known as 7 Darter Info Services Pvt. Ltd.)" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. The report presents the work and learning undertaken during my internship from 15 June 2026 to 27 July 2026. I affirm that this report has not been submitted to any other institute for any degree or diploma requirement and that I shall be responsible for any copyright violation in this regard.

Signature: _____

Student Name: ANJALI

Roll No: 2410030568

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Naadyx Solutions Private Limited for providing me the opportunity to undertake the summer internship as a Software Development Intern. The internship provided practical exposure to software development, computer vision, video processing and problem solving in a real-world technical environment.

I am thankful to the professionals and team members who provided guidance, feedback and technical support during the development work. Their practical insights helped me understand how software solutions are designed, implemented, tested and improved to address real-time requirements.

I also express my gratitude to IILM University, School of Computer Science and Engineering, for providing the academic framework and support required to complete the internship and prepare this report.

Signature: _____

Student Name: ANJALI

Roll No: 2410030568

INTERNSHIP COMPLETION CERTIFICATE

NAADYX SOLUTIONS PRIVATE LIMITED
(Formerly Known As : 7 Darter Info Services Pvt. Ltd.)
CIN : U62099DL2023PTC410834 | www.7darter.com
Unit No 1305, 13th Floor, Vasavi Sky City
Gachibowli Cir, Hyderabad, Telangana 500032
Phone: +91 9571463074
Email: navdeepdagur@7darter.com



Certificate of Internship

Date: 30/07/2026

Anjali,
IILM University, Greater Noida

This certifies that **Ms. Anjali** has completed an internship at Naadyx Solutions Private Limited as a **Software Development** intern. She worked with the team from **June 15th to July 27th, 2026**, specializing in providing cutting-edge solutions, consultancy/advisory services, and products in Defense and security, Space and aviation, Cyber Security Network Systems, Wireless Network Systems, and Software Development.

She actively contributed to various projects, analyzing complex problem statements and providing innovative solutions. Working alongside experienced professionals, she gained practical knowledge and hands-on experience in addressing real-time problem statements across multiple sectors. Her teamwork, communication, and problem-solving skills were good.

We wish you all the best in your future endeavors.

7 DARTER INFO SERVICES-PVT. LTD.

Navdeep Dagur

Navdeep Dagur
Director



Director
Naadyx Solutions Private Limited

Registered Address : B-19, Phase-1, Okhla Industrial Estate Phase-1, New Delhi, South East
Delhi, Delhi, 110020

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PROJECT DESCRIPTION

4.1 Introduction

During the internship, I worked as a Software Development Intern on a Flask-based web application for video-based vehicle number plate detection and recognition. The application allows a user to upload a video, processes the video using OpenCV and EasyOCR, detects number plates, highlights detected plates in the output video, validates recognized plate strings using a project-specific regular-expression pattern, and identifies the most common valid number plate detected in the video.

The project provided practical exposure to Python web development, computer vision, optical character recognition, video processing, file handling and debugging. The solution was intended to reduce dependence on manual vehicle identification and make plate information available from video footage in a more systematic manner.

4.2 Organization Profile

Naadyx Solutions Private Limited, formerly known as 7 Darter Info Services Pvt. Ltd., is an information-technology company focused on cutting-edge solutions, consultancy/advisory services and products for real-time problem statements. Its stated areas include Defense and security, Space and aviation, Cyber Security Network Systems, Wireless Network Systems and Software Development.

The company works with skilled experts and developers having practical experience in the relevant technical sectors. The internship certificate confirms that the internship was completed as a Software Development Intern from 15 June to 27 July 2026.

4.3 Problem Statement

Manual identification and tracking of vehicles from CCTV or recorded video can be time-consuming and difficult to scale. The project addressed this problem by developing a web-based system capable of accepting video input, detecting candidate number-plate regions, extracting text using OCR, validating the recognized text, and presenting the processed video with detected plates highlighted.

The project-specific validation logic used the regular-expression pattern `^[A-Z0-9]{7}$`, which accepts seven uppercase alphanumeric characters. This pattern was used as a validation rule within the application rather than as a universal representation of every vehicle registration format.

4.4 Project Objectives

1. Develop a Flask-based web application for uploading and processing video files.
2. Use OpenCV-based image and video-processing techniques to locate candidate number-plate regions.

3. Use EasyOCR to extract alphanumeric text from detected plate regions.
4. Validate OCR results using a predefined regular-expression pattern.
5. Highlight detected number plates in the processed video output.
6. Determine the most common valid number plate detected in the processed video.
7. Provide processed video output for preview and download.
8. Gain practical experience in Python, Flask, OpenCV, OCR, video processing and application debugging.

4.5 Scope of the Project

- Video-file upload through a web interface.
- Frame-by-frame video processing using OpenCV.
- Contour-based candidate plate detection and image preprocessing.
- Canny edge detection as part of the computer-vision pipeline.
- Text extraction using EasyOCR.
- Validation of recognized plate strings using a configurable regex pattern.
- Generation of an output video with detected plates highlighted.
- Identification of the most frequently detected valid plate in the processed video.
- Secure handling of uploaded and processed video files within the application.

4.6 Technologies and Tools Used

Category	Technology / Tool	Purpose
Frontend	HTML5, CSS3	User interface for uploading videos and viewing results.
Backend	Python, Flask	Web server, routing, file handling and application logic.
Computer Vision	OpenCV	Video processing, contour detection, image preprocessing and frame handling.
OCR	EasyOCR	Recognition of characters from detected number-plate regions.
Validation	Python Regular Expressions	Validation of recognized plate strings.
Development	Python virtual environment, pip	Dependency and project environment management.
Version Control	Git / GitHub	Source-code management and project versioning.

4.7 System Architecture

The application follows a web-based processing architecture. The user interacts with the Flask web interface to upload a video. Flask receives the file and passes it to the video-processing pipeline. OpenCV reads the video and processes frames, while image-processing operations are used to identify candidate plate regions. EasyOCR extracts text from the candidate regions. The extracted text is then checked against the configured validation pattern. Valid detections are highlighted in the output video, and the application identifies the most common valid plate detected.

System Flow

User	Flask Upload	OpenCV Frame Processing	Plate Detection + EasyOCR	Regex Validation	Processed Video + Result
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At a high level: User → Flask upload interface → OpenCV frame processing → contour/edge-based candidate detection → EasyOCR text extraction → regex validation → highlighted output video and most common valid plate.

4.8 Methodology

- 1. Requirement Analysis:** Understanding the requirement for automated number-plate detection from uploaded video and defining the expected input and output.
- 2. Project Setup:** Creating the Flask application structure, virtual environment and required dependencies.
- 3. Video Upload:** Accepting video files through the web interface and storing them for processing.
- 4. Frame Processing:** Reading video frames with OpenCV and applying image-processing operations.
- 5. Plate Detection:** Using contour detection and Canny edge detection to identify candidate regions that may contain number plates.
- 6. OCR:** Passing candidate plate regions to EasyOCR to extract alphanumeric text.
- 7. Validation:** Applying the project-specific regex pattern `^[A-Z0-9]{7}$` to filter recognized strings.
- 8. Output Generation:** Drawing/highlighting detected plates and producing the processed video for preview and download.
- 9. Result Selection:** Identifying the most common valid plate detected across the processed video.
- 10. Testing and Debugging:** Running the application, checking output, resolving processing issues and refining detection performance.

4.9 Expected Outcomes and Results

- A functional Flask-based application capable of accepting video uploads and processing them.
- Automatic detection of candidate vehicle number-plate regions in video frames.
- OCR-based extraction of plate characters using EasyOCR.
- Validation of recognized plate strings using the configured regular-expression rule.
- A processed video in which detected plates are highlighted.
- Identification of the most common valid plate detected in the input video.
- Practical understanding of integrating a Python backend with computer-vision and OCR libraries.
- The project description provided for the internship states that frame skipping and predefined license-plate patterns were used to improve processing performance, with detection speed reported as increasing by 30% and accuracy by 25%.

4.10 Certificate of Completion and Supporting Documentation

The internship completion certificate issued by Naadyx Solutions Private Limited is attached in this report. The certificate states that Ms. Anjali completed the internship as a Software Development Intern from June 15th to July 27th, 2026. It also identifies the company's work across Defense and security, Space and aviation, Cyber Security Network Systems, Wireless Network Systems and Software Development.

The complete certificate is reproduced on the following page for institutional record.

INTERNSHIP COMPLETION CERTIFICATE

NAADYX SOLUTIONS PRIVATE LIMITED
(Formerly Known As : 7 Darter Info Services Pvt. Ltd.)
CIN : U62099DL2023PTC410834 | www.7darter.com
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Delhi, Delhi, 110020

5. BIBLIOGRAPHY / REFERENCES

1. Naadyx Solutions Private Limited / 7 Darter Info Services Pvt. Ltd., company information supplied for the internship documentation, www.7darter.com.
2. 7 Darter Info Services Private Limited, company profile and posts supplied through LinkedIn: <https://www.linkedin.com/company/7darter/posts/?feedView=all>
3. Project repository supplied for the internship project:
<https://github.com/username/repo-name.git>
4. Python documentation and Flask/OpenCV/EasyOCR documentation as used for the relevant software-development concepts.